TI-58C: Changes and Improvements compared to TI-58/59

The TI-58C is a TI-58 with constant memory: its programs and data survive power-off. Its firmware is a revision of the TI-59/TI-58 firmware, not a new design. This document describes what the TI-58C ROM does differently, and why.

Addresses are TI-58C ROM addresses unless written "TI-59 XXXX". They refer to the annotated TI-58C ROM listing in the Calc-U-59 repository. Statements marked *derived* come from reading the code; nothing observed shows them directly.

Overview

	TI-59 / TI-58	**TI-58C**
ROM chips	TMC0582, TMC0583, TMC0571B	CD2400, CD2401 (TMC0580), TMC0573
Date code on the dumped chips	1979	1980
User RAM	serial RAM (TMC0598), RAM_OP bus word	CMOS constant memory, `MEMWR` / `MEMRD`, 64 raw registers
Registers	120 (TI-58: 60)	60 user registers + 4 extra cells E000–E003
Default partition	479.59 (TI-58: 239.29)	239.29
Card reader	yes (TI-58: code present, no reader)	no, code removed
Printer-present line	KP at digit 0	KP at digit 10
Idle instructions	¼ speed	full speed

How much is shared.

- 4,953 of the 6,144 ROM words align with the TI-59 in order. Another 505 words are TI-59 code that was moved, word for word apart from the jump targets.
- The remaining ~690 words are new or rewritten. Most of them are the former card-reader area 1680–17FF.
- The arithmetic, the transcendental functions, the display formatter, AOS, the program scanner and the print mnemonic table are all unchanged.

The 16 numeric constants and all keycode sub-programs except two are byte-identical. The two that differ are bug fixes; see below.

Why there are more differences than the hardware explains

On paper only three things change: a different RAM chip, a different clock chip, and no card reader. Even so, about a fifth of the ROM differs. These causes compound:

1. **Every RAM access site had to be rewritten.**

- The TI-59 addresses its serial RAM with one bus word. It carries the operation in nibble 0 and a BCD register number, plus a hexadecimal tens nibble for registers ≥ 100. Block operations, such as clearing ten registers, are a single word.
- `MEMWR` / `MEMRD` take a *binary* address (Sout[1]·16 + Sout[0]) and move one register at a time.
- So every place that touches RAM gained a BCD → binary conversion: data registers, program registers, LRN store, Ins/Del, label search, CMs/CP, Op 09 and Op 17.

2. **Branch range.** The new code is longer, so routines no longer fit where they were.
   - Whole TI-59 routines were moved to free space: PGM_SELECT, Op 16/17, the return-stack push, the tan kernel, HIR, INV List, the library label search, and others.
   - Jumps that now cross too far go through one- or two-word trampolines. The spare words of the print-mnemonic table hold many of them.
3. **The printer-present line moved to digit 10.**
   - The idle loop's key scans see digit 10 as keyboard row 10, so an attached printer looks like a held key. The idle loop was rebuilt around this (see "Idle loop").
   - The trace gate, List, Adv, Prt and the new Op 40 all test the line at the new digit.
4. **The card-reader code went away.** That freed most of 1680–17FF, which now holds a factory self-test and relocated routines.
5. **The ROM is a later revision.**
   - Its dumped chips carry 1980 date codes; the TI-59 set carries 1979.
   - TI used the revision to fix six firmware bugs, two of them in the keycode sub-programs, and to add Op 40 and a TRACE marker.
6. **Fix and the partition persist.** Constant memory keeps them across power-off. Fix and Op 17 therefore write them into RAM, and power-on restores them.

Constant memory

Addressing

- `MEMWR`: the instruction sends the address on the bus, and the IO value of the *next* instruction is written, within that instruction's field. If the next instruction drives no IO, zero is written. CMs, CP and the first-power-up clear use this to clear registers.
- `MEMRD` puts the register on the bus for the next load (`D=LOAD`, `A=LOAD`).
- **Data register nn:**
  - REG_ADDR (`0D15`) computes the RAM index 59 − nn in BCD, as the TI-59 does with its own top register.
  - RAM_ADDR (`0D96`) then counts that index down in BCD while it counts A[1..0] up in binary.
  - RAM_ACCESS (`0A1A`) dispatches to read (`0D9E`), write (`0DFE`) or the LRN path.
  - Ind is decided after a normal read (`0DA3 → 0F5D`).
- **Program register n** is raw register n:
  - fetch `0B7D–0B85 → 0EDA`
  - label search `0BDC–0BE3 → 0F74`
  - LRN write-back `0C16–0C1F`
  - Ins/Del and Op 09 write `1224–122E → 0F6A`

- **Cost:** the count loop makes low register numbers slowest. RCL 00 takes 691 active instructions, RCL 29 takes 594; on the TI-59 any RCL takes about 466.

E000: the system register

Raw register 60 (0x3C), called E000 here, holds a signature plus the two settings that survive power-off:

Nibbles	Content	Written at
15–3	signature: the mantissa of constant 0, ln 10 = 2302585092994	first power-up, `037E-0380`
2–1	partition ÷ 10 (BCD)	Op 17, `17AE-17B5`
0	Fix setting (n + 2, 0 = floating)	Fix / INV Fix, `0474-0481`

The address is built from SCOM[9] nibble 1 and 0xC. On the TI-58C, SCOM[9] nibble 1 is the high nibble of E000's address (3). On the TI-59 it counts RAM chips. E001–E003 (raw 61–63) are only touched by the first-power-up clear and the self-test.

Power-on (`0350-03A6`)

Steps from `0000` to the idle loop:

1. **Clear SCOM.** `0000` clears all 16 SCOM rows, as on the TI-59. SCOM loses its contents at power-off, so this runs on every power-on.
2. **Settle delay.** `0352-0355` count a 51-pass loop of about 100 active cycles before the first RAM access. It has no TI-59 counterpart. *Derived* purpose: to let the CMOS RAM interface settle.
3. **RAM-size probe.** `035C-0369` write the signature value to raw register 0x7F and read it back.
   - A 64-register RAM reads back 0. E000 is then 0x3C, and the clear loop makes 64 passes.
   - A 128-register RAM keeps the value. E000 is then 0x7C, the partition becomes 6, the top register 119, and the clear loop makes 128 passes.
   - So the ROM also supports a TI-59-sized CMOS RAM: with a patched emulator it starts up as 479.59, and STO 99 works. This replaces the TI-59's RAM_OP probe of register 90.
4. **Signature check.** `036A-036F` compare nibbles 3–15 of E000 with the ln 10 mantissa. Nibbles 0–2 are not validated (*derived*).
5. **First power-up.** If the signature does not match (new batteries, or a corrupted E000):
   - `0370-0379` write zero to all 64 raw registers, E000–E003 included.
   - `037A-0380` write E000 with the signature, partition 3 and Fix floating.
6. **Restore.** `0381-03A6` rebuild the system SCOM rows from E000:
   - SCOM[9]: partition, and the E000 address nibble.
   - SCOM[13]: partition and top register 059.
   - SCOM[0]: Fix.

   Then the code falls into CLR.

The TI-58C does not back up the rest of SCOM. There are no `MEMWR` / `MEMRD` instructions in the idle or pre-idle path; the only keys that write RAM are Fix, STO-type operations and LRN stores. A power

cycle traced after `Fix 2, Rad, St flg 3, Eng, 4 Op 17`, a program, an LRN position, `Pgm 02` and a pending `7 + 3 (` gives:

Survives	Lost
program and data registers	angle unit (back to DEG)
partition	Eng/Sci
Fix	user flags
	pending AOS operations and parentheses
	t, X
	step pointer (LRN shows 000)
	return stack
	selected module program

Idle loop and timing

Printer line. With a printer attached, the KP line at digit 10 is active all the time. The rebuilt idle loop (`062A-0659`, plus `0A25-0A30`) handles this:

- The release gate masks KP at rows 10, 11, 13 and 14. It runs its window twice, counted by a new toggle fA[11].
- The arming and confirmation scans start at row 12. Rows 10 and 11 go to `0A25`, which rescans rows 9–1 and then re-enters the confirmation scan.
- The TI-59 card-switch poll (TI-59 `063C-0640`) is gone.

Clock. On the TI-59 an instruction in the idle state takes four times as long as an active one; on the TI-58C idle instructions run at full speed. The ROM compensates in one place only:

Activity	TI-59	TI-58C	Compensated?
Error blink	4 counts per lit pass, both phases ≈ 450 ms with printer: period 0.90 s; stand-alone 0.45 s (both measured on a real TI-59)	1 count per pass, 100 + 100 passes ≈ 400 + 533 ms: period 0.933 s (0.920 s measured on a real TI-58C with a 389.4 kHz clock, which the model gives at that clock); with printer 333 + 400 ms	yes: the lit-phase +3 was dropped; the period stays ≈ 0.9 s
Steady idle pass	9.0 ms (18.0 ms with printer)	4.0 ms (6.7 ms with printer)	—
Key debounce gap (one digit sweep)	4.5 ms	1.33 ms	no
Pause (DISPLAY_PEEK `04E2`, unchanged)	432 ms	128 ms	no
GTO peek while running	216 ms	64 ms	no

The TI-59 blink and Pause figures were checked against a real TI-59 filmed next to a stopwatch. On the TI-59 the printer-present line sits at D0, which its idle scan sees as a "row 0" key and discards: two extra digit sweeps per pass with a printer. The TI-58C times are computed from the emulator's model: 384 kHz ÷ 32 = 12,000 instructions per second (the TI-58's part, assumed), idle at full speed. Real units vary: one measured PHI 194.7 kHz (389.4 kHz) and blinks 0.920 s, another blinks 0.896 s (as at 400 kHz); each unit's other measurements are compared at the clock its blink gives. The short Pause is real: a filmed TI-58C (the 0.896 s unit) shows each Pause lit for about 0.122 s; the model gives 0.118–0.120 s at that unit's clock.

Bug fixes

Each fix can be reproduced in Calc-U-59 by switching the model between TI-59 and TI-58C: the TI-59 shows the bug, the TI-58C does not.

#	TI-58C	TI-59	Bug on the TI-59	Example
1	`03E3` CLR fA[7]	`03DB`	An Ind pending from run mode survives into LRN. The first recorded key gets an extra 00 step.	`Ind LRN 5` stores 00 05 (TI-58C: 05); `Ind LRN STO 12` stores 00 42 12
2	`0920` CLR fA[7]	`092E`	`Ind List` starts one step late and garbles operands.	program STO 12 lists "001 12 B" instead of "000 42 STO / 001 12 12"
3	`0A60–0A62`	`0A5B–0A5E`	A keyboard Σ+, x̄, D.MS, P→R or Op 11–15 pushes the step counter unchanged. After a halt or SST (program-register cache valid, fB[11] clear) that counter names the step last executed, so the next R/S runs that step again.	program `1 2 3 4 5 R/S`, keys `RST SST SST Σ+ R/S` show 2345 (TI-58C: 345)
4	`0D30` CLR fA[10]	—	With an Ind pointer register outside the partition, the pending trace record (fA[10]) suppresses the "?" error line, so the error goes untraced. Affects STO/RCL/SUM/Prd/Exc/Fix/Op/GTO/SBR/Pgm Ind.	printer TRACE on: `5 STO Ind 65` prints no "5. ?" line
5	`12E3` MOV R5,fB[1..4]	`11BD` (fA)	HIR SUM/Prd/INV SUM/INV Prd take the operand's exponent sign from the display flags (fA[3]). These are stale in a running program and clear in a non-sci display, so 0 < |x| < 1 is used as m·10^+|e|. HIR only reaches a RAM program	`3 HIR 08 .5 HIR 48 HIR 18` gives 150 (TI-58C: 1.5)

#	TI-58C	TI-59	Bug on the TI-59	Example
			through pseudo-codes, the Ind merge or cards. The keycode sub-programs are not affected.	
6	keycode step 031	step 031	Op 12/14/15: a CE right after the slope clears the error of a 0/0 or x/0 slope (all x equal, or one data point). The TI-58C has CP there instead, which only clears t (overwritten by the next x⇄t), so the error reaches the result.	`2 x⇄t 1 Σ+ 2 x⇄t 3 Σ+ 0p 12` shows a steady 0 / t = 1 (TI-58C: flashing)
7	keycode steps 251–281	251–283	R⟶P (INV P⟶R) ends with a CE meant for the deliberate 0/0 of the quadrant trick at x = 0. It also hides an overflow of $x^2 + y^2$. The TI-58C computes θ first, clears, then computes r, and no longer clobbers Hir 7.	`3 x⇄t 2 EE 50 INV P→R` gives r = 1E50 without an error (TI-58C: flashing)

Fix 7 has a side effect: an underflowing $x^2 + y^2$ (for example y = 2E-60) now flashes, although r is correct. The new R⟶P also has an extra CE at step 252 with no observable effect.

Enhancements

- **Op 40** (`08D8 → 0BFA`) sets flag 7 if a printer is connected. Otherwise flag 7 is left unchanged, and INV has no effect. On the TI-59 every Op 40–99 is an error; on the TI-58C, Op 41–99 still are.
- **TRACE marks taken branches** (`15F0 → 17D4-17E9 → 14EB`). When x=t, x≥t or If flg transfers, the address line prints "▲" (PC-100 code 61) before the address, so `1 x≥t 006` prints "▲6".
  - Branches not taken are printed without the marker.
  - Dsz, Ind operands, GTO and SBR never get it.
  - The test reads the continuation slot in SR (0x0C0, 0x0BC, 0x0B8) and fB[8].
- **Factory self-test.**
  - **Entry:** the idle loop polls test lines (`064B`: KN at digit 15; KN on row 8 or 9), which a test fixture drives.
  - **Sequencer** (`0DB7-0DDA`): prints the test numbers 1–11 and dispatches through the table `0C01-0C0A`. Test numbers 3 and 7 have no body.

Test	Body	Checks
1	`09CC`	constant-ROM entry 7
2	`08BB`	constant-ROM entry 63
4, 5	`16C7`, `16C8`	SCOM rows 0–7 and 8–15, with a 6969…/9696… checkerboard

Test	Body	Checks
6	`1680`	all 64 raw RAM registers, with the same checkerboard; prints "D" first on a 128-register RAM
8, 9	`1360`, `1368`	SCOM rows written with their own index
10	`1375`	RAM[k] := k for all 64 registers

  - **Result:** pass goes `10BF → 0DC3`; fail goes `10BE → 0DD6 → 09DD`, which ends in a cold restart.
  - **Side effect:** the test wipes SCOM and all of constant memory, E000 included.
  - **Open:** a KR line at digit 10 (`064E → 09F7`) jumps to 0x1800, beyond the 6K ROM. Possibly a hook for test-fixture code; unexplained.
- **Fix and the partition survive power-off** (see "Constant memory").

Kept from the TI-59 (still there)

- **Fast mode.** The St flg / KEY_ENCODE mis-dispatch at `0B34` and the HIR STO into SCOM[0] (`11BC`) work exactly as on the TI-59 (TI-59 Fast Mode). The 69! test program runs in 363k weighted cycles in fast mode against 585k normally.
- **The 7th SBR level is still dropped.** The push moved to `1334`, and the depth check is at `1339-133C`.
- **Unchanged behaviour:**
  - the Pgm, library label search and library transfer handling;
  - Op 09, and Op 09's refusal of protected modules;
  - LIB_RETURN, label search, Dsz, CMs/CP semantics, List, INV List and Adv.
- `CRD.OFF` (`0005`) is still issued at reset, with no effect.

Removed

- **The card reader:**
  - TI-59 `1680-17AB` (CARD_IO);
  - the card gate and Write entry (TI-59 `0459-0461`);
  - the idle-loop card-switch poll.

  Write (slot `0069`) now only redisplays X (`5 Write` shows 5.).
- **Memory protection,** which on the TI-59 a protected card sets in SCOM[9] nibble 2. Its tests are gone from LRN, List, the trace gate, the GTO peek, Op 17 and CP.
- **The RAM_OP bus word.** Its block clears are replaced by loops.